

Subject Description Form

Subject Code	ME6021 - ME6023
Subject Title	Special Topics in Mechanical Engineering I/II/III
Credit Value	3
Level	6
Pre-requisite/ Co-requisite/ Exclusion	Knowledge of Materials and Mechanics equivalent to the final year of an Honours Degree in Mechanical Engineering course. Preference will be given to those who has had several years experience in the topic chosen.
Objectives	This subject aims to equip students with the comprehensive knowledge in a selected research topic from the following areas in Mechanical Engineering: advanced materials, advanced energy source, computational fluid mechanics, combustion science, vehicle pollution, acoustics and waves.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand the state-of-the-art developments and trends in a selected research topic from the following areas: advanced materials, advanced energy source, computational fluid mechanics, combustion science, and vehicle pollution after-treatment, acoustics and waves. b. Familiarize themselves with the technical knowhow and the tools for the analysis and design made in the selected research topic.
Subject Synopsis/ Indicative Syllabus	A. Advanced materials A1. Introduction to advanced materials and technology: mechanics of composites and smart materials, product design process issues, limitations and constraints, Environment-friendly product design process, and application case studies. A2. Introduction to micro-technology and nano-technology: Concepts, principles, physical, mechanical and thermal properties at nano and microscales, scanning probe microscopy, SEM, TEM, nanofluids, carbon nano-materials, nanofabrication and manufacturing. B. Advanced energy source B1. Clean energy generation and their applications: principles and properties of various hydrogen storage/conversion systems for the new generation of vehicle design, bio-energy in waste water/solids treatment, alternative energy source for modern vehicles B2. Principles of electric vehicles: design requirements, power supply systems, air/noise pollution

	C. Computational fluid mechanics C1. Computational fluid dynamics and application in heat transfer: theory and methods; applications in product design and engineering systems C2. Numerical methods in thermofluid dynamics: Physical modeling, numerical methods, thermofluid dynamics problems D. Combustion science D1. Fundamentals of combustion science: nature of combustion, fuels and alternative fuels, thermodynamics and kinetics of combustion, combustion-led air pollution D2. Current trend in combustion for environmental protection: sulfur-free liquid fuels, low-NOX combustion, bio-fuels, hydrogen, hydrocarbon gaseous fuels enriched with hydrogen E. Vehicle pollution after-treatment E1. Introduction to vehicular pollution: problems and their feasible solutions, formation and destruction of vehicular pollutants, measurement of vehicular pollutants, particulate trap, diesel oxidation catalyst and 3-way catalytic converter, performance of particulate trap, diesel oxidation catalyst and 3-way catalytic converter E2. Applications: cyclone and electrostatic precipitator, after-treatment devices for nitrogen oxides, fibrous filters, vehicular exhaust after-treatment systems E3. Vibration/acoustics: Advanced topics related to sound and vibration; Vibroacoustic analysis; Numerical modeling methods including FEM/BEM; Nonlinear dynamics; Passive and active control. F. Acoustics, waves, and vibration F1. Advanced topics related to sound and vibration; Vibroacoustic analysis; Numerical modelling methods including FEM/BEM; Nonlinear dynamics; Passive and active sound control. F2. Wave dispersion; Waves in layered plates and hollow cylinders; waves in viscoelastic medium. Bulk waves in weakly nonlinear elastic medium; second harmonic generation.
Teaching/Learning Methodology	The subject can be conducted via guided study in two modes for individual students. <u>Mode I</u> requires a student to take an MSc subject related to the topics of the guided study subject or a relevant short course as the basis of the guided study subject. The student will be required to participate fully in the MSc subject/relevant short course (i.e. attend all the lectures, complete both the coursework and examination requirements). To bring the subject up to the doctoral level, a student is required to submit further write-ups and presentations. An overall grade for the guided study subject is then derived from the result of the MSc subject as well as the extra writes-up and presentations. <u>Mode II</u> is operated for guided study subjects with no relevant MSc subject/short course available. A student is required, under the supervision of the subject

	supervisor, to read specified monographs, journal publications and/or a book. The student and the subject supervisor must meet regularly to discuss the progress made by the student in the subject. Coursework in terms of literature survey reports and presentations should normally be included. At the end of the semester the student will be examined, normally both orally and in written form. All of the above contributes to both intended learning outcomes of the subject. <u>Alignment of learning & teaching activities with the ILOs (Mode I)</u> <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="2">Intended subject learning outcomes</th></tr><tr><th>a</th><th>b</th></tr><tr><td>1. Lectures</td><td>✓</td><td>✓</td></tr><tr><td>2. Tutorials / Laboratories</td><td>✓</td><td>✓</td></tr><tr><td>3. Literature survey</td><td>✓</td><td>✓</td></tr><tr><td>4. Write-ups and presentations</td><td>✓</td><td>✓</td></tr></table> <u>Alignment of learning & teaching activities with the ILOs (Mode II)</u> <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="2">Intended subject learning outcomes</th></tr><tr><th>a</th><th>b</th></tr><tr><td>1. Literature survey</td><td>✓</td><td>✓</td></tr><tr><td>2. Write-ups and presentations</td><td>✓</td><td>✓</td></tr></table>	Teaching/Learning Methodology	Intended subject learning outcomes		a	b	1. Lectures	✓	✓	2. Tutorials / Laboratories	✓	✓	3. Literature survey	✓	✓	4. Write-ups and presentations	✓	✓	Teaching/Learning Methodology	Intended subject learning outcomes		a	b	1. Literature survey	✓	✓	2. Write-ups and presentations	✓	✓
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Assessment Methods in Alignment with Intended Learning Outcomes	For Model I study, it includes the courseworks and examination of an MSc subject and additional assignments given by the teaching staff. The additional assignments include write-ups and presentations. <table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="2">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th></tr><tr><td>1. MSc subject (include coursework and examination)</td><td>67</td><td>✓</td><td>✓</td></tr><tr><td>2. Additional assignments (include write-ups and presentations)</td><td>33</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100</td><td colspan="2"></td></tr></table> For Mode II study, the Coursework part will include the coursework of the study assigned by the teaching staff. The Examination part will include the written and oral examination of the study assigned by the teaching staff.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		a	b	1. MSc subject (include coursework and examination)	67	✓	✓	2. Additional assignments (include write-ups and presentations)	33	✓	✓	Total	100												
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	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed	
			a	b
	1. Coursework (normally assignments and presentations)	45	✓	✓
	2. Examination (normally both written and oral, conducted by the responsible staff and a staff member who is knowledgeable in the topic)	55	✓	✓
	Total	100		
Student Study Effort Expected (Mode I)	Class contact (time-tabled):			
	▪ Lecture			24 Hrs.
	▪ Tutorial/Laboratory/Practical Classes			15 Hrs.
	Guided activities:			
	▪ Meeting with the supervisor / Presentations/ Viva examination			10 Hrs.
	▪ Self-study / Preparation of reports and presentation materials			56 Hrs.
	Total student study effort			105 Hrs.
(Mode II)	Guided activities:			
	▪ Meeting with the supervisor / Presentations/ Viva examination			20 Hrs.
	▪ Self-study / Preparation of reports and presentation materials			85 Hrs.
	Total student study effort			105 Hrs.
Reading List and References	Will be assigned by the teaching staff.			
Last Update	June 2024			